Survey Design Seminar
Survey Methods 670
Winter 2026

Week 8: Ethical issues in data collection, data storage, data management, and report writing

February 26, 2026

Outline of Today's Lecture

A. Human Subjects
B. Data collection, storage and dissemination
C. Copyright
D. Collaboration
E. Peer review, authorship and publication
F. Use of Artificial Intelligence/Large Language Models
G. Case studies

A. Human Subjects

- Standards for conduct of research
  - ~~Fabrication~~
  - ~~Falsification~~
  - ~~Plagiarism~~
- Researcher's obligations to participants, clients, society and research community
  - Respondents: beneficence, justice, respect for persons, safeguarding confidentiality. IRB approval
  - What to provide to the public (AAPOR code): sponsor, questions asked, population and sample frame, methods of data collection, sample size, precision of findings.

Readings:

Groves, R. M., Fowler, F. J., Couper, M. P., Lepkowski, J. M., Singer, E., & Tourangeau, R. (2009). Chapter 11 – Principles and Practices Related to Ethical Research. Survey Methodology (2nd ed.). Hoboken, New Jersey: Wiley Series in Survey Methodology.

Oakes, J. M. (2002). Risks and Wrongs in Social Science Research An Evaluator's Guide to the IRB. Evaluation Review, 26(5), 443–479.

B. Data collection, storage and dissemination

- Standards in data collection and storage
  - Total Survey Error
    - Coverage error, sampling error, non-response error
    - Validity, measurement error, processing error
  - Measures to protect privacy and confidentiality
    - Data that are subject to privacy restrictions must be stored in a safe place that is accessible only to authorized personnel.
    - Use random codes to identify individual subjects, rather than names or social security numbers,
    - The researcher who collects or uses the information has the primary responsibility for its protection.

Readings:

National Academy of Science. 2009. On Being a Scientist: Responsible Conduct of Research, Second Edition, Washington, D. C.: National Academies Press, "Treatment of Data".

B. Data collection, storage and dissemination

- Data dissemination
  - Fast
  - Accessibility and Replicability
    - Researchers can withhold data until they have had time to establish their priority for their work through publication. Withholding data prior to publication is a commonly accepted practice that most researchers and funding agencies accept.
    - Once a researcher has published the results of an experiment, it is generally expected that all the information about that experiment, including the final data, should be freely available for other researchers to check and use.
    - Some journals formally require that the data published in articles be available to other researchers upon request or stored in public databases.

Readings:

National Academy of Science. 2009. On Being a Scientist: Responsible Conduct of Research, Second Edition, Washington, D. C.: National Academies Press, “Treatment of Data”.

C. Copyright

- Business of idea generation
- Protecting intellectual property
  - Patent: exclusive use for a defined period (15-20 years) as long as owner provides full description of manufacture, use, and function.
  - Copyright: protects words and images, not underlying ideas (70-95 years)
- Interference with publishing
  - Most journals require signing away copyright for publication
    - Author's Addendum: legal instrument that modifies the publisher's agreement and allows you to keep certain rights to your article
  - Recent changes to NIH require all funded research to be made publicly-available on NIH website upon publication.

D. Collaboration

- <u>Collaboration</u> – the action of working with someone to produce or create something (i.e. sharing research techniques, multi-centered surveys in different countries)

- <u>Competition</u> – rivalry between two or more persons or groups for an object desired in common (i.e. researchers competing for the same grant)

D. Collaboration: Case Scenario

Andy, Sunghee, & Kristen met during a discussion at a professional meeting.

They share a common interest in survey incentives but come from different scientific backgrounds.

The trio sees benefits from working together & starts sketching out a grant proposal.

The scientific ideas seem to fit together well, but it is not clear how they should handle some logistics …

D. Collaboration: Preliminary Questions

- Who should submit the proposal -- through which university?
- Do all three need to get IRB approval to work on the project?
- What will happen if their work has practical applications?

D. Collaboration: Communication

- Collaborators should:
  - Share findings with colleagues in the collaboration
  - Pay attention to what others are doing
  - Deliberately set aside time to discuss problems and findings
  - Communicate any important changes in your work, such as changes in key personnel
  - Share news and developments
    - Everyone in the collaboration should be equally knowledgeable about important project information

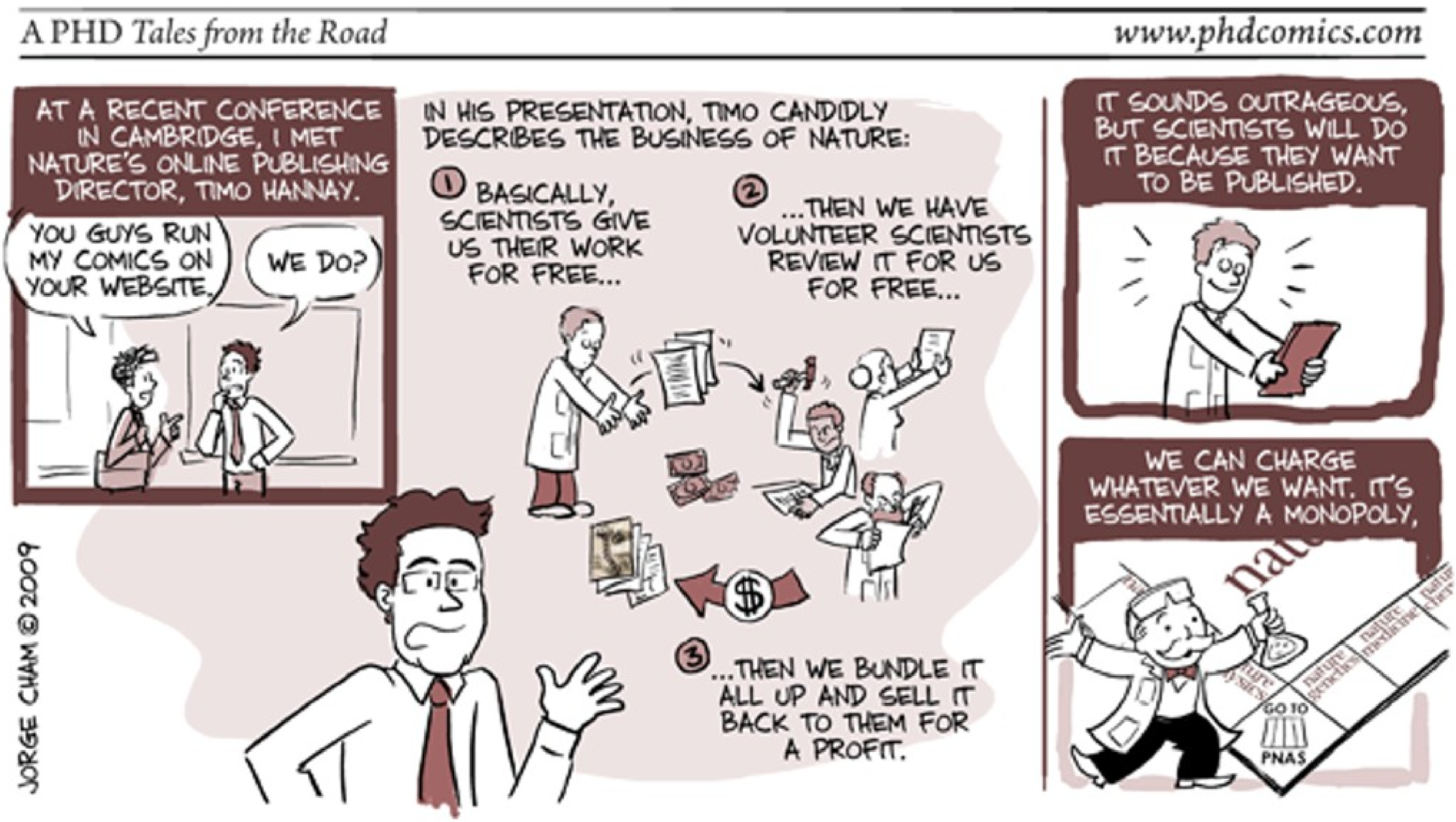
D. Collaboration: Before Starting

- What do we expect to get out of this work?
- Who is going to do what and when?
- How will data be collected, stored, & shared?
- Who will have access to the data after we’re done?
- When does the collaboration end?

D. Collaboration: Before Starting

- Who will give presentations & decide content?
- Who is going to draft publications?
- How will we decide authorship?
- When will we publish?
- Who owns the intellectual property?

E. Peer review, authorship and publication



E. Peer review, authorship and publication

- Authorship and allocation of credit
  - Can vary by field
  - Lead author does bulk of writing
  - Last author can be senior: genesis of idea but carried out by others
  - Avoid “guest” and “ghost” authorship

Readings:

National Academy of Science. 2009. On Being a Scientist: Responsible Conduct of Research, Second Edition, Washington, D. C.: National Academies Press, “Authorship and the Allocation of Credit”

E. Peer review, authorship and publication

- Authorship scenario
  - You are a graduate student who works in Dr. Smith’s lab. You have made a significant discovery and feel that you deserve first authorship of the publication; however, Dr. Smith suggests that the first paper announcing the results be published under his name to give it broader circulation. What should you do?

E. Peer review, authorship and publication

- 1665: Henry Oldenburg, secretary of the Royal Society of London, started the practice of sending submitted manuscripts to experts who could judge their quality before publication.
- Peer reviews: as an author
  - Your reviewers are a reflection of your readers: if they misunderstand something, others probably will also.
  - Don't always have to do what is requested, but must explain why you are not, or why you are doing something differently.
- Peer reviews: as a reviewer
  - If you publish, you need to review!
  - What makes a good review
    - Timely
    - Don't feel compelled to criticize
    - Actionable suggestions
- Committee on Publishing Ethics (COPE)

Readings:

Starbuck, W. H. (2003). Turning Lemons Into Lemonade Where Is the Value in Peer Reviews? Journal of Management Inquiry, 12(4), 344–351.

E. Peer review, authorship and publication

- Peer review scenario
  - Dr. Sung is struggling with the decision whether to agree to review the work of a graduate student at another university for publication in the major journal of her field because she is concerned about a possible conflict of interest. The student's work overlaps with her research in many ways, which is one of the reasons she has been asked to serve as a reviewer. And Dr. Sung wants to avoid charges that she unfairly used some of the student's ideas when she publishes her own work. Should Dr. Sung agree to do the review? Why or why not?

F. Use of Artificial Intelligence/Large Language Models

- Explosion of AI/LLMs starting in 2023 has added a new issue about ethical behavior
- Bifurcation:
  - Use of neural networks in advanced statistical methods for prediction models – Ok, new "tool in the toolbox".
    - Still many open issues: incorporating uncertainty
  - Use of LLMs to produce written material, generate code, conduct other tasks – more complicated issue.
    - Developing closed-ended codes for open-ended questions seems to have no ethical issues
    - Code and summarizing papers for personal use OK but potential issues about errors "hallucinations"
    - Writing text for publication and generating "fake" data out of bounds.
  - Clear guidance is still lacking.

G. Case study: "When contact changes minds: An experiment on transmission of support for gay equality" by Michael J. LaCour and Donald P. Green

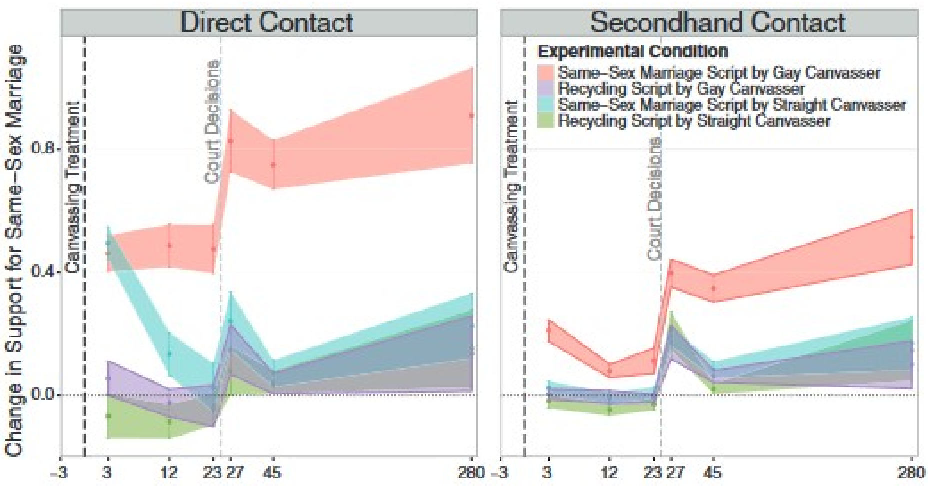
- Falsify (social research) data
- Reporting findings
- Senior researcher's role
- Retraction
- Consequences

G. Case study: "When contact changes minds: An experiment on transmission of support for gay equality" by Michael J. LaCour and Donald P. Green

LaCour, Michael J. & Green, Donald P. (2014). When contact changes minds: An experiment on transmission of support for gay equality. Science 346(6215): 1366-1369.

- Randomized trial comparing using gay (n=22) vs. straight (n=19) canvassers to encourage support for same-sex marriage in Southern California households.
- Found large initial effects on both groups, but only consistent changes in households contacted by gay canvassers

→ Contact with minorities + discussion of pertinent issues can produce opinion changes



G. Case study: “When contact changes minds: An experiment on transmission of support for gay equality” by Michael J. LaCour and Donald P. Green

Broockman, David, Kalla, Joshua, and Aronow, Peter (19 May 2015). "Irregularities in LaCour (2014)" (PDF).

- Tried to replicate but found much lower response rates
  - Survey firm that had supposedly conducted study had no knowledge of study.
- Found that baseline outcome data matched data from a national survey and over-time changes that are unusually small and indistinguishable from perfectly normally distributed noise.
- Suggest that the baseline data were generated from a random sample of the 2012 Cooperative Campaign Analysis Project (CCAP) and normally distributed noise was added to simulate follow-up waves.

G. Case study: “When contact changes minds: An experiment on transmission of support for gay equality” by Michael J. LaCour and Donald P. Green

Gelman, Andrew (May 20, 2015). "Fake study on changing attitudes: Sometimes a claim that is too good to be true, isn’t". The Washington Post.

- Enormous effect sizes
- Small errors → responses are too consistent
- “Ironically, LaCour benefited (in the short term) by his strategy of completely faking it. If he’d done the usual strategy of taking real data and stretching out the interpretation, I and others would’ve been all over him for overinterpreting his results, garden of forking paths, etc. But, by doing the Big Lie, he bypassed all those statistical concerns.”

G. Case study: “When contact changes minds: An experiment on transmission of support for gay equality” by Michael J. LaCour and Donald P. Green

Singal, Jesse. "Is Social Science a Giant Liberal Conspiracy?", New York Magazine (June 8, 2015).

- Accepted by Science because of its scientific value, or “because it `flattered the ideological sensibilities of liberals’”? (Wall Street Journal editorial)
- Or accepted because it was “novel” – ran counter to prior research showing it is difficult to change political views
- Damages trust in science

G. Case study: “When contact changes minds: An experiment on transmission of support for gay equality” by Michael J. LaCour and Donald P. Green

Foster, Drew. "Will Academia Waste the Michael LaCour Scandal?", New York Magazine (June 5, 2015)

- Did the system work?
- System is too competitive
- (Social science) data is too easy to falsify
- Green added his name without meaningful participation in the study

G. Case study: “Artificial Intelligence, Scientific Discovery, and Product Innovation” by Adrian Toner-Rodgers

2024 Arxiv paper abstract: https://arxiv.org/abs/2412.17866

This paper studies the impact of artificial intelligence on innovation, exploiting the randomized introduction of a new materials discovery technology to 1,018 scientists in the R&D lab of a large U.S. firm. AI-assisted researchers discover 44% more materials, resulting in a 39% increase in patent filings and a 17% rise in downstream product innovation. These compounds possess more novel chemical structures and lead to more radical inventions. However, the technology has strikingly disparate effects across the productivity distribution: while the bottom third of scientists see little benefit, the output of top researchers nearly doubles. Investigating the mechanisms behind these results, I show that AI automates 57% of “idea-generation” tasks, reallocating researchers to the new task of evaluating model-produced candidate materials. Top scientists leverage their domain knowledge to prioritize promising AI suggestions, while others waste significant resources testing false positives. Together, these findings demonstrate the potential of AI-augmented research and highlight the complementarity between algorithms and expertise in the innovative process. Survey evidence reveals that these gains come at a cost, however, as 82% of scientists report reduced satisfaction with their work due to decreased creativity and skill underutilization.

G. Case study: "Artificial Intelligence, Scientific Discovery, and Product Innovation" by Adrian Toner-Rodgers

Problem: this paper by a second year PhD student at MIT was entirely made up (including liberal use of AI to generate fake data!)

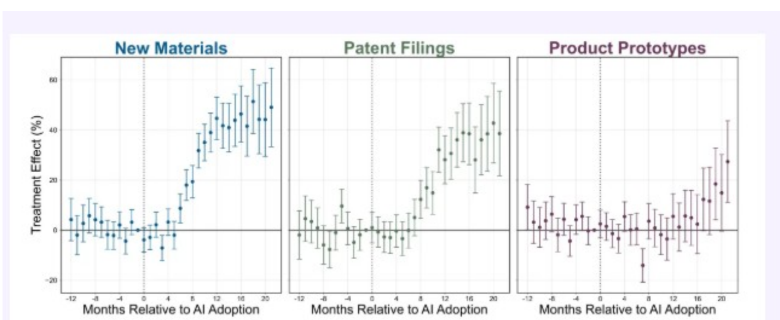
- Was on its way to acceptance at the Quarterly Journal of Economics (top economics journal).
- Received coverage at the Wall Street Journal, The Atlantic, and Nature.

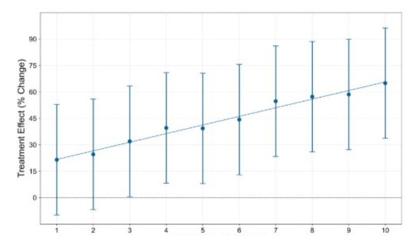
G. Case study: “Artificial Intelligence, Scientific Discovery, and Product Innovation” by Adrian Toner-Rodgers

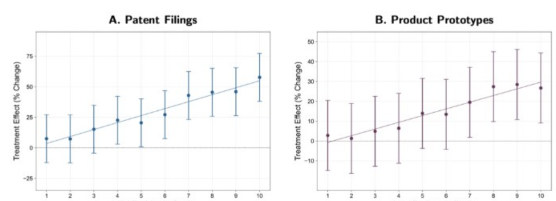
(From Andrew Gelman’s blog https://statmodeling.stat.columbia.edu/ and https://thebsdetector.substack.com/p/ai-materials-and-fraud-oh-my)

Discovered by a materials scientist who wanted to find out more about the details of the study:

- Very few companies have this many materials scientists and none have the even distribution of subspecialties claimed in the paper.
- Very clean findings:







G. Case study: “Artificial Intelligence, Scientific Discovery, and Product Innovation” by Adrian Toner-Rodgers

- Real research is addressing this issue, and generally finds that AI is most adept at helping the weakest performers – the opposite of the Toner-Rogers “findings”.
- Real world impact is serious: the European Central Bank cited this paper as a argument for targeting billions of euros for investment.
  - Has 93 citations include 10 as of 2026. Nearly all treat the paper as legitimate.
- All of this is not helping a crisis in public confidence about science.